



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded.
Department of Civil Engineering ISE-II Examination Academic Year: 2025-26 (Semester I)

Class: F.Y. M. Tech. (W.M.)
Date: 21/11/2025

Course: Groundwater Engg.
Time: 04:30 P.M. - 05:15 P.M

Course Code: PCC- WM503
Max. Marks: 20

Note: i. Assume suitable data whenever necessary and state it clearly. ii. Mobile phones and programmable calculators are strictly prohibited. iii. Exchange/Sharing of stationery, calculator etc. not allowed. iv. Figures to right indicate full marks. v. The symbols carry usual meaning.

Q. No.	Question	Marks	CO	BT
Q.1	What are the assumptions in Theim's equation? A well penetrates fully into a confined aquifer and is pumped at a constant rate of $3000 \text{ m}^3/\text{d}$. If the transmissivity of the aquifer is $1000 \text{ m}^2/\text{d}$. The piezometric head at an observation well 100 m away from the production well is 50 m. Find the piezometric head at another observation well located 700 m from the production well axis.	10	CO3	BT3
Q.2	Derive the discharge-drawdown equation for a steady radial flow towards a well in an unconfined aquifer.	10	CO4	BT2



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering
ISE-2 Examination
Academic Year: 2025-26

Class: FY-M.Tech(AI)

Subject: Statistical Machine Learning

Subject Code: PCC-AI503

Date: 21.11.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iii. All questions are compulsory.

Que. No.	Questions	Marks	CO	BT
Q.1	Describe the step-by-step working of the K-Means algorithm with a neat diagram.	6	CO2	BT1
Q.2	Explain the Logistic Regression algorithm in detail. Write its cost function and perform optimization on cost function.	6	CO3	BT1, BT2
Q.3	Explain the Gradient Descent optimization algorithm in detail.	4	CO1	BT2
Q.4	Write a difference between parametric and Non parametric ML models.	4	CO2	BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT):

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes: On successful completion of this course, students should be able to:

CO 1	Identify appropriate statistical learning methods for the given problem involving real data.
CO 2	Identify other possible problems with messy data, such as multicollinearity, understand their consequences, and propose solutions.
CO 3	Use training and testing data to evaluate performance of the chosen regression and classification techniques and compare them.
CO 4	Apply cross-validation techniques to find the optimal degree of flexibility-the best subset of predictors or the optimal tuning parameters.
CO 5	Show, analytically or empirically, the optimal balance between precision within training data and prediction power.
CO6	Develop theory of generalization.
CO7	Develop recommendation system.



**SHRI GURU GOBIND SINGHJI INSTITUTE OF ENGINEERING &
TECHNOLOGY,VISHNUPURI,NANDED**
Department Of Computer Science & Engineering
First Year M. Tech(CNIS)
In-Semester Examination-II (Year 2025-2026)

Date:21//11/2025	Time:-04:30-05:15	Subject:-Advanced Operating Systems (PCC-NS503)	Marks-20
-------------------------	--------------------------	--	-----------------

	Questions	Marks	CO(BT)
Q.1	What is Distributed Mutual Exclusion? Explain any two distributed mutual exclusion algorithms (e.g., Lamport's Algorithm, Ricart-Agrawala Algorithm).	05	CO2 (BT1,BT2)
Q.2	Compare Distributed OS and Network OS. Explain at least five differences with examples of each system.	05	CO3(BT2)
Q.3	Write short notes on: 1. Remote Procedure Call (RPC) 2. Peer-to-Peer Networks	05 05	CO2(BT4) CO3(BT4)

******* All the Best *******

CO's	Statement
CO1	Understand the fundamental and advances of operating systems and its security and protection mechanisms.
CO2	Learn the fundamentals of distributed operating systems and real time operating systems.
CO3	Apply the concept of network operating system, RTOS and related issues.
CO4	Analyze differences between traditional and advanced operating systems in the context of scheduling of processes, memory management and file system management.
CO5	Design and implement different operating system program components and compare them.
BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create	



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science and Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination II

Class: MTech CNIS First Year	Information Security	Course Code: PCC-NS517
Date: 19/11/2025	Time: 4.30 to 5.15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc, not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Describe the major steps involved in the contingency planning process.	5	CO5	BT 4
	b	Explain the concept of "spheres of security" in designing security architecture.	5	CO4	BT 2
		OR			
	b	Describe how biometric recognition is used to verify identity.	5	CO4	BT3
Q.2	a	Describe the five goals of information security governance.	5	CO3	BT 2
	b	Differentiate between Mandatory Access Control (MAC) and Discretionary Access Control (DAC) based on 4 factors controlling authority, access decision, flexibility and security level.	5	CO2	BT 4

== ALL THE BEST ==

CO1 :Define the concepts of and definition of information system
CO2:Learn several types of information system.
CO3:Apply threats to information security
CO4:Analyze how to protect information resources
CO5:Design and develop maintaining and protecting information system.

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation II

Class: First Year M. Tech.	Course Name: CAD/CAM	Course Code: PCC-DM 502
Date: 19.11.2025	Time: 16:30 to 17:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	A rectangle is formed by four points whose coordinates are: A(50, 50), B(100, 50), C(100, 80), and D(50, 80). Determine the coordinates of four points for new rectangle in reduced size using the scaling factors 0.5 and 0.6 along X and Y-directions respectively.	8	CO3	BT4/ BT2
Q.2	Explain constructive solid geometry (CSG).	6	CO4	BT2
Q.3	What is sketched base modeling? Explain the steps used in creating models using sketched features	6	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Identify the key components of a CAD/CAM system and understand the concept of product life cycle, hardware & software, modules and applications.
CO 2	Apply design principles and computer applications to solve real-world design problems by constructing 3D models, assemblies and generative drawings.
CO 3	Apply the concept of transformation for generating different positions of given problem with defined geometry.
CO 4	Use CAD software to create and modify solid and surface models, including complex shapes and assemblies.
CO 5	Understand the principles and applications of MRP-I and MRP-II in manufacturing systems and use AI and simulation techniques to optimize processes.
CO 6	Design and develop CNC programs using G and M code for complex geometry using CAM software and apply CAD data exchange formats.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: M. Tech. (AI) First Year	Subject: Digital Image and Video Processing	Subject Code: PCC-AI-502
Date: 19/11/2025	Time: 16:30 to 17:15	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Discuss the theoretical foundations and application areas of Walsh, Hadamard, and Haar transforms. How do these transforms differ in their approach to representing image data?	5	CO1	BT1
Q.2	Explain the Discrete Fourier Transform (DFT) in two dimensions. Derive its mathematical representation, discuss its properties, and highlight its applications in image processing.	5	CO3	BT3
Q.3	Discuss Histogram Equalization and Histogram Specification (Matching) as methods of image enhancement. Explain their algorithms and importance.	5	CO2	BT2
Q.4	Describe Frequency Domain Filtering for image enhancement. Explain the process involving Fourier transform, filtering, and inverse transform. Discuss commonly used frequency filters.	5	CO2	BT1

— ALL THE BEST —

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create
Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand theory and models in Image and Video Processing
CO 2	Interpret and analyze 2D signals in frequency domain through image transforms.
CO 3	Apply quantitative models of image and video processing for various engineering applications.
CO 4	Develop innovative design for practical applications of image and video processing in various fields.



Shri Gurn Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Information Technology
Academic Year: 2025-26 (semester I)
In-Semester Examination II

Class: FY M.Tech.	Course Name: Advanced Database Management System	Course Code: PCC-IT503
Date: 21.11.2025	Time: 16:30 to 17:15 pm	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Question	Marks	CO	BT
Q.1	Explain Transaction System Process using suitable diagram?	10 M	CO2	BT5
Q.2	Describe Concurrency Control with suitable example?	10 M	CO3	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	introduce the basics of Database Administration.
CO2	give a detailed understanding of how to maintain a database quickly & accurately.
CO3	design and manage the database server to solve the issues related to the database server.
CO4	introduce the basics of database administration.
CO5	to give a detailed understanding of how to maintain a database quickly & accurately.
CO6	design and manage the Database Server to solve the issues related to the Database Server.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class:
FY M.Tech.
(CAD/CAM)

Subject: Computing Tools and
Programming

Subject Code: PCC-DM503

Date: 21.11.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. Attempt all the questions.

Q.N.		Marks	CO	BT
	Solve the following questions.			
Q.1	Write a C program and detailed algorithm to find the largest and smallest element in an array using pointers and functions.	08	CO1	BT6
Q.2	Write a C program and detailed algorithm to sort an array in ascending order (Bubble Sort).	08	CO1	BT6
Q.3	Classify and write different types of SQL Commands. (Write atleast 2 from each category and mention the category as well.)	04	CO4	BT1,3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1:	Write programs in languages like C, C++, C#, VB.
CO4:	Explain database management concepts, DBMS structures, relational models, and apply SQL commands for data definition, manipulation, querying, and reporting.



Shri Guru Gobind Singh Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination II
Academic Year: 2024-25 (Semester – I)

Class:
FY M.Tech.

Subject: Computer Aided Design

Subject Code: PEC-PL504

Date: 19-11-2024

Time: 16:30 TO 17:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Question	Marks	CO	BT
1	Discuss the different tools in sketching and modelling.	5	CO2	BT3
2	Discuss different data exchange tools in any CAD package	5	CO3	BT1
3	Write a short on geometric transformation	5	CO4	BT2
4	Discuss reflection geometric transformation with help of example	5	CO1	BT3

— ALL THE BEST —

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Classify CAD hardware and software for variety of applications.
CO 2	Exhibits importance and use of various CAD modelling techniques.
CO 3	Exhibits the idea and procedure of data transfer in various CAD tools.
CO 4	Calculate the position, shape and geometry of CAD model after translation or transformation
CO5	Demonstrate the assembly procedure of CAD applications.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

In-Semester -02-Examination (Semester-I) 2025-26

M.Tech Structures

Behavior of Concrete Structures

Civil Engineering

PCC-SE502

Max. Marks: 20

Date: 19/11/2025

Time: 16:30-17:15

- Note: i. All questions are compulsory.
ii. Assume suitable data if necessary
iii. Use of non-programmable calculator is permitted.
iv. This paper contains 1 page.

Q.N.	Question	Marks	CO	BT
Q.1	a Compression test on 5 samples of 150 mm cubes of concrete gave following results: 26MPa, 27 MPa, 31 MPa, 33 MPa, and 29Mpa. What is the characteristic strength of concrete.	04	CO 2	BT3
	b What are the different types of shear failure? Explain with neat sketch.	03	CO 1	BT1
Q.2	a Find the load carrying capacity of axially loaded column of size 230x525mm, for minimum percentage of steel. Use M25 and Fe415 grade of concrete and steel resp.	05	CO 3	BT4,5
	b Design the shear reinforcement for a beam at its shear span. The size 230x450mm subjected to 14kN/m of superimposed dead load and 8kN/m of live load. The beam is reinforced with 2-16+2-12 Φ at bottom and 3-16 at top. The length of the beam is 5m.	08	CO 4	BT4

IS 456 : 2000

Table 19 Design Shear Strength of Concrete, τ_c , N/mm²
(Clauses 40.2.1, 40.2.2, 40.3, 40.4, 40.5.3, 41.3.2, 41.3.3 and 41.4.3)

$100 \frac{A_s}{bd}$	Concrete Grade					
	M 15	M 20	M 25	M 30	M 35	M 40 and above
(1)	(2)	(3)	(4)	(5)	(6)	(7)
≤ 0.15	0.28	0.28	0.29	0.29	0.29	0.30
0.25	0.35	0.36	0.36	0.37	0.37	0.38
0.50	0.46	0.48	0.49	0.50	0.50	0.51
0.75	0.54	0.56	0.57	0.59	0.59	0.60
1.00	0.60	0.62	0.64	0.66	0.67	0.68
1.25	0.64	0.67	0.70	0.71	0.73	0.74
1.50	0.68	0.72	0.74	0.76	0.78	0.79
1.75	0.71	0.75	0.78	0.80	0.82	0.84
2.00	0.71	0.79	0.82	0.84	0.86	0.88
2.25	0.71	0.81	0.85	0.88	0.90	0.92
2.50	0.71	0.82	0.88	0.91	0.93	0.95
2.75	0.71	0.82	0.90	0.94	0.96	0.98
3.00 and above	0.71	0.82	0.92	0.96	0.99	1.01



Shri Guro Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination II

Class: FY M.Tech.

Course Name: Surface Water Hydrology

Course Code: PCC-WM502

Date: 19.11.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT														
Q.1	a	Explain the factors affecting infiltration capacity.	05	CO1	BT2														
	b	The mass curve of rainfall of duration 100 minutes is given below. The catchment had an initial loss of 0.6 cm. The ϕ- index of the catchment is known to be 0.6 cm/hour. Calculate the total surface runoff from the catchment due to this storm. <table><tr><td>Time from start (min)</td><td>0</td><td>20</td><td>40</td><td>60</td><td>80</td><td>100</td></tr><tr><td>Cumulative rainfall (cm)</td><td>0</td><td>0.5</td><td>1.2</td><td>2.6</td><td>3.3</td><td>3.5</td></tr></table>	Time from start (min)	0	20	40	60	80	100	Cumulative rainfall (cm)	0	0.5	1.2	2.6	3.3	3.5	05	CO2	BT3
Time from start (min)	0	20	40	60	80	100													
Cumulative rainfall (cm)	0	0.5	1.2	2.6	3.3	3.5													
Q.2	a	What is hydrograph? Explain the different sections of hydrograph.	04	CO3	BT2														
	b	Characteristics of two catchments M and N measured from a map are given below: <table><tr><td>Item</td><td>Catchment M</td><td>Catchment N</td></tr><tr><td>L_{ca}</td><td>76 km</td><td>52 km</td></tr><tr><td>L</td><td>148 km</td><td>106 km</td></tr><tr><td>A</td><td>2718 km²</td><td>1400 km²</td></tr></table> For the 6-h unit hydrograph in catchment M, the peak discharge is at 200m³/s and occurs at 37 h from the start of the rainfall excess. Assuming the catchment M and N are meteorologically similar, determine the elements of the 6-h synthetic unit hydrograph for the catchment N by using Snyder's method.	Item	Catchment M	Catchment N	L_{ca}	76 km	52 km	L	148 km	106 km	A	2718 km ²	1400 km ²	06	CO3	BT4		
Item	Catchment M	Catchment N																	
L_{ca}	76 km	52 km																	
L	148 km	106 km																	
A	2718 km ²	1400 km ²																	

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To train students in surface water hydrology as a prerequisite for water resource management
CO 2	To impart in-depth knowledge of hydrologic processes, data collection and management, and estimation of parameters.
CO 3	To develop students for application of concepts and methods of hydrology to field problems.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Vishnupuri, Nanded- 431606
Department of Electronics and Telecommunication Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)**

Class:
F. Y. M. Tech(AI)
Date: 17/11/2025

Subject: Artificial Neural Network and Applications
Time: 16:30-17:15

Subject Code: PCC-AI 501
Max. Marks: 20

Note:

- Use of non-programmable calculator is permitted
- Assume suitable data if necessary

			Marks	COs	BT
Q 1		Explain K-means clustering with its limitations. Apply the K-means algorithm on following training samples $x_1 = (0,1,0,1)$; $x_2 = (0,1,1,1)$; $x_3 = (0,0,0,0)$; $x_4 = (1,0,0,1)$ Use the initial centroid values for two centroids as, $C_1 = (0.2,0.6,0.6,0.3)$ and $C_2 = (0.5,0.6,0.3,0.3)$. Perform only one iteration. Also, calculate the errors present in the calculations.	10	CO6	4,5
Q 2	(a)	What is a radial basis function? What is the necessity of a radial basis function? Explain Gaussian function.	05	CO5	2
	(b)	What is bias and variance in neural networks? Explain with a suitable example. What is the relationship between bias, variance, and overfitting? List the different techniques used to overcome the problem of overfitting in artificial neural networks.	05	CO3	2,3

Bloom's Taxonomy (BT)	Course Outcomes	On successful completion of this course, students will be able to	tasks such as
1- Remember,	1.	Understanding the basic structures of artificial neural network, their limitations, basic pattern analysis	classification and clustering, learning and adaptation using the learning rules, Implementation of learning rule.
2- Understand,	2.	Describe the concepts of feed forward neural networks using single layer and multilayer networks to solve classification problem, and its implementation, single layer feedback networks to study the concept of memory using neural networks.	
3- Apply,	3.	Analyze and implement the applications of neural networks in character recognition and control systems.	
4- Analyse,	4.	Understand Auto associative neural networks, Pattern storage and retrieval, Hopfield model, recurrent neural networks	
5- Evaluate,	5.	Analyze Bayesian neural networks, Radial basis function networks	
6- Create	6.	Understand self-organizing maps and recent trends in neural networks	



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
ISE-III Examination (Semester - II)
Academic Year: 2025-26

Class: FY-M.Tech.
(INFT/PLM)

Subject: Project Management

Subject Code: AUD-901

Date: 19/11/2025

Time: 16:30 pm to 17:15 pm

Max. Marks: 20

Note:

- i. Non-programmable calculator is permitted.
- ii. Figures to right indicate full marks.

Q.No.	Question	Marks	CO	BT																																																		
Q1	Draw network & Find out the time required to complete the following project and the critical activities: <table><thead><tr><th>Activity</th><th>Predecessor Activity</th><th>Optimistic time estimate (to days)</th><th>Most likely time estimate (tm days)</th><th>Pessimistic time estimate (tp days)</th></tr></thead><tbody><tr><td>A</td><td>-</td><td>2</td><td>4</td><td>6</td></tr><tr><td>B</td><td>A</td><td>3</td><td>6</td><td>9</td></tr><tr><td>C</td><td>A</td><td>8</td><td>10</td><td>12</td></tr><tr><td>D</td><td>B</td><td>9</td><td>12</td><td>15</td></tr><tr><td>E</td><td>C</td><td>8</td><td>9</td><td>10</td></tr><tr><td>F</td><td>D, E</td><td>16</td><td>21</td><td>26</td></tr><tr><td>G</td><td>D, E</td><td>19</td><td>22</td><td>25</td></tr><tr><td>H</td><td>F</td><td>2</td><td>5</td><td>8</td></tr><tr><td>I</td><td>G</td><td>1</td><td>3</td><td>5</td></tr></tbody></table> Also calculate variance and standard deviation of the project.	Activity	Predecessor Activity	Optimistic time estimate (to days)	Most likely time estimate (tm days)	Pessimistic time estimate (tp days)	A	-	2	4	6	B	A	3	6	9	C	A	8	10	12	D	B	9	12	15	E	C	8	9	10	F	D, E	16	21	26	G	D, E	19	22	25	H	F	2	5	8	I	G	1	3	5	10	CO2	BT3
Activity	Predecessor Activity	Optimistic time estimate (to days)	Most likely time estimate (tm days)	Pessimistic time estimate (tp days)																																																		
A	-	2	4	6																																																		
B	A	3	6	9																																																		
C	A	8	10	12																																																		
D	B	9	12	15																																																		
E	C	8	9	10																																																		
F	D, E	16	21	26																																																		
G	D, E	19	22	25																																																		
H	F	2	5	8																																																		
I	G	1	3	5																																																		
Q2	a Define: EST, LST, EFT & LFT.	03	CO2	BT1																																																		
	b Which are the Different Tools & Techniques for Project Management. Differentiate between CPM and PERT.	07	CO2	BT2																																																		

== ALL THE BEST ==

Page 01 of 02

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO 1	Understand the concepts and functions of project management.
CO 2	Apply the project plan planning and monitoring techniques
CO 3	Analyze the project value, risk and quality.
CO 4	Design and develop projects at each stage of the software development life cycle (SDLC).



Class: FY MTech.

Course Name: Finite Element Analysis of Structures

Course Code: PCC-SE 501

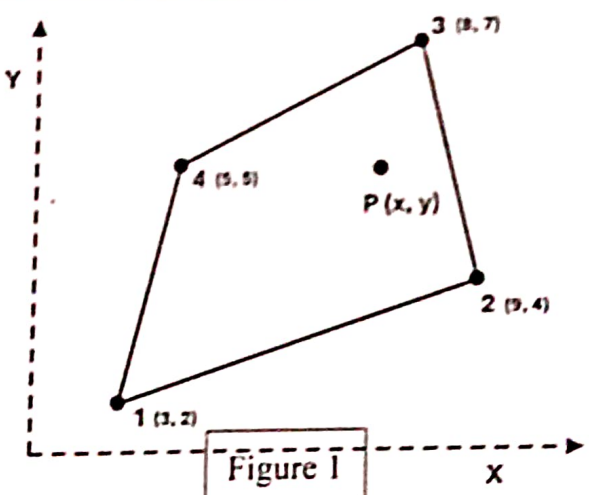
Date: 17.11.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Statement	Marks	CO	BT
Q.1 a	Derive and explain the shape function for a 2-noded beam element using the Cartesian coordinate system.	06	CO4	BT4
Q.1 b	Derive shape functions for a four noded rectangular element in natural coordinate system using Lagrange interpolation function.	04	CO5	BT3
Q.2 a	Determine the Cartesian coordinates (x, y) of a point 'P' as shown in Figure 1. The natural coordinates of 'P' are (0.45, 0.64).  Figure 1	06	CO4	BT4
Q.2 b	Write and explain with example the types of iso parametric elements.	04	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To recall the fundamentals of engineering mathematics and theory of structures.
CO 2	To understand the energy principles, finite element concept, stress analysis, meshing, nonlinear problems and its applications.
CO 3	To apply basic concept of finite element method to solve solid mechanics problems.
CO 4	Analyze and provide approximate solution to any finite element problems.
CO5	Evaluate the performance of one dimensional and two-dimensional problems.
CO6	To recall the fundamentals of engineering mathematics and theory of structures.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Information Technology
In-Semester Examination II
Academic Year: 2025-26 (Semester – I / II)

Class: M.Tech
Date: 19.11.2025

Subject: Machine Learning
Time: 16:30 to 17:15

Subject Code: PCC-IT502
Max. Marks: 20

Q.N.		Marks	CO	BT
Q.1	Given the following data points: (2, 3), (3, 4), (5, 6), (8, 8), (10, 10) Perform the first two iterations of the K-Means clustering algorithm manually with k=2. Assume initial centroids as (2,3) and (8,8).	10	CO2,5	BT1-5
Q.2	A Discuss how unsupervised learning is applied in recommendation systems.	5	CO2,4	BT1-6
	B Explain the difference between hard clustering and soft clustering.	5	CO1	BT1-2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Demonstrate a Strong Grasp of Machine Learning Concepts
CO 2	Effectively Preprocess and Analyse Data
CO 3	Select and Evaluate Machine Learning Models
CO 4	Apply Machine Learning Techniques to Real-World Problems
CO 5	Understand and Address Ethical Issues in Machine Learning